



Introduction to Computer Networks (SWE3022)

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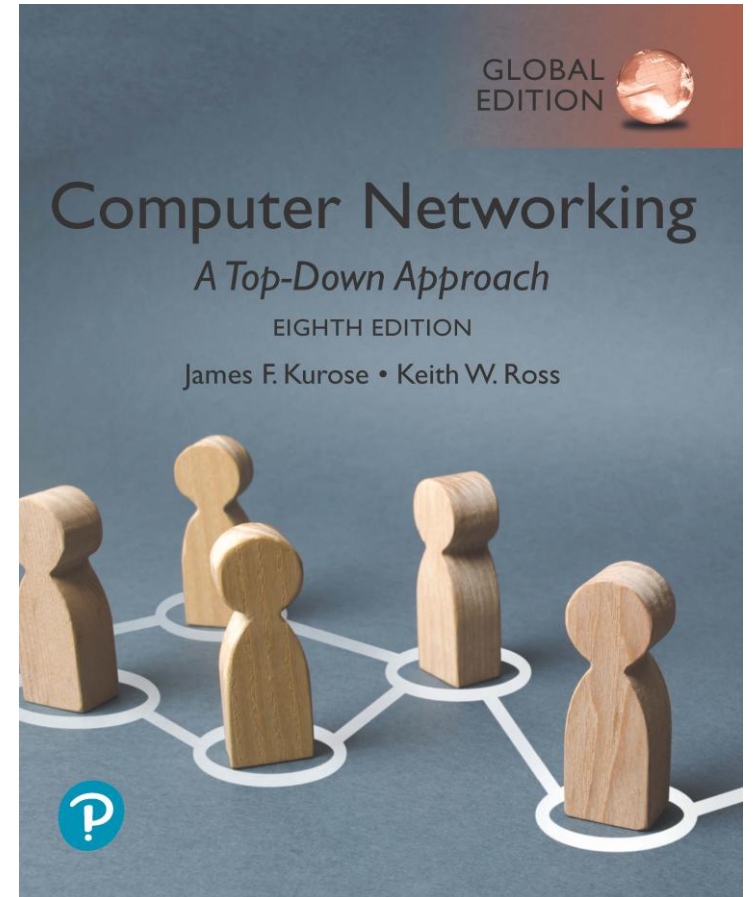
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Chapter 7. Wireless and Mobile Networks: Part 2



Computer Networking: A Top-Down Approach

8th Edition, Global Edition

Jim Kurose, Keith Ross

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Chapter 7 outline

- Introduction

Wireless

- Wireless links and network characteristics
- WiFi: 802.11 wireless LANs
- Cellular networks: 4G and 5G



Mobility

- Mobility management: principles
- Mobility management: practice
 - 4G/5G networks
 - Mobile IP
- Mobility: impact on higher-layer protocols

4G/5G cellular networks

- *the* solution for wide-area mobile Internet
- widespread deployment/use:
 - more mobile-broadband-connected devices than fixed-broadband-connected devices (5-1 in 2019)!
 - 4G availability: 97% of time in Korea (90% in US)
- transmission rates up to 100's Mbps
- technical standards: 3rd Generation Partnership Project (3GPP)
 - www.3gpp.org
 - 4G: Long-Term Evolution (LTE) standard

4G/5G cellular networks

similarities to wired Internet

- edge/core distinction, but both below to same carrier
- global cellular network: a network of networks
- widespread use of protocols we've studied: HTTP, DNS, TCP, UDP, IP, NAT, separation of data/control planes, SDN, Ethernet, tunneling
- interconnected to wired Internet

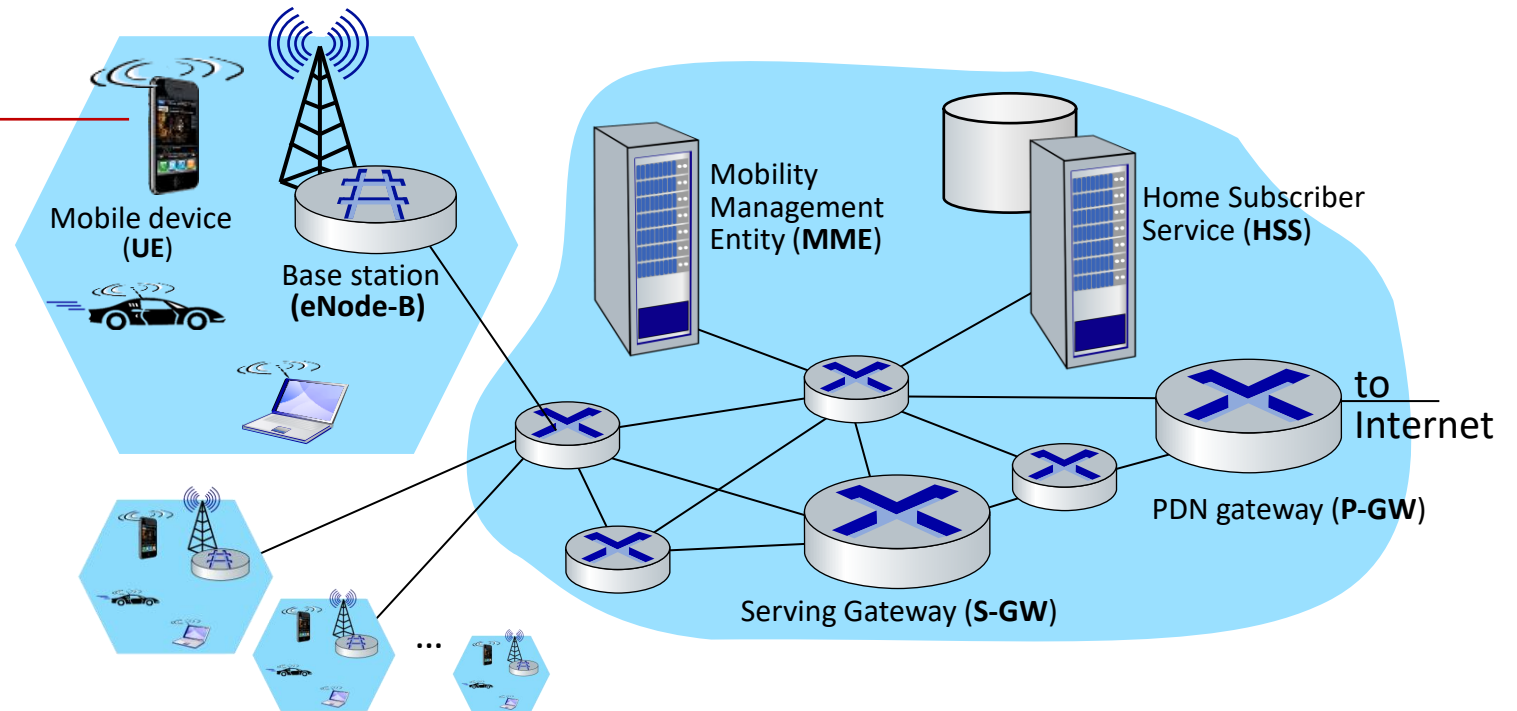
differences from wired Internet

- different wireless link layer
- mobility as a 1st class service
- user “identity” (via SIM card)
 - SIM: Subscriber Identification Module
- business model: users subscribe to a cellular provider
 - strong notion of “home network” versus roaming on visited nets
 - global access, with authentication infrastructure, and inter-carrier settlements

Elements of 4G LTE architecture

Mobile Device:

- smartphone, tablet, laptop, IoT, ... with 4G LTE radio
- 64-bit International Mobile Subscriber Identity (IMSI), stored on SIM (Subscriber Identity Module) card
- LTE jargon: User Equipment (UE)

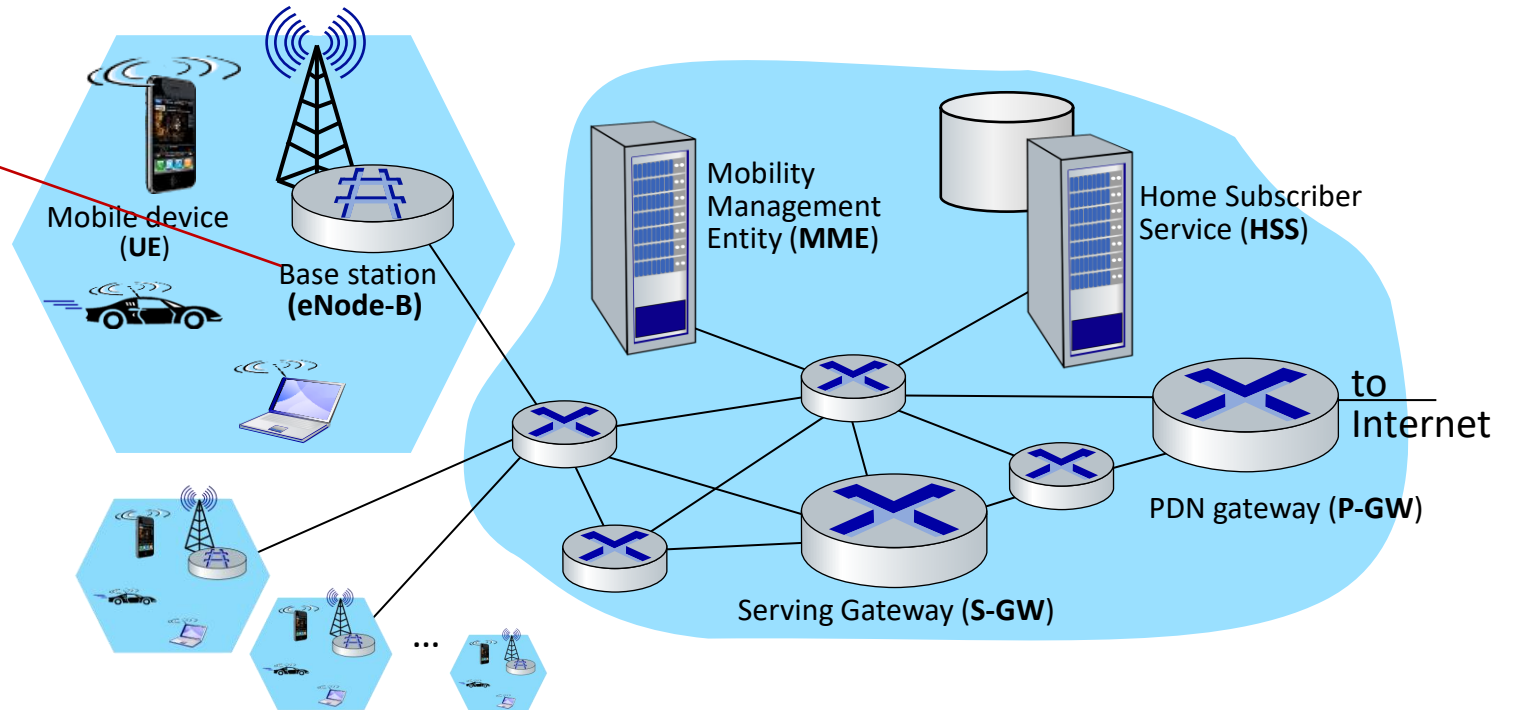


← Radio Access Network (RAN) → ← **all-IP** Enhanced Packet Core (EPC) →

Elements of 4G LTE architecture

Base Station:

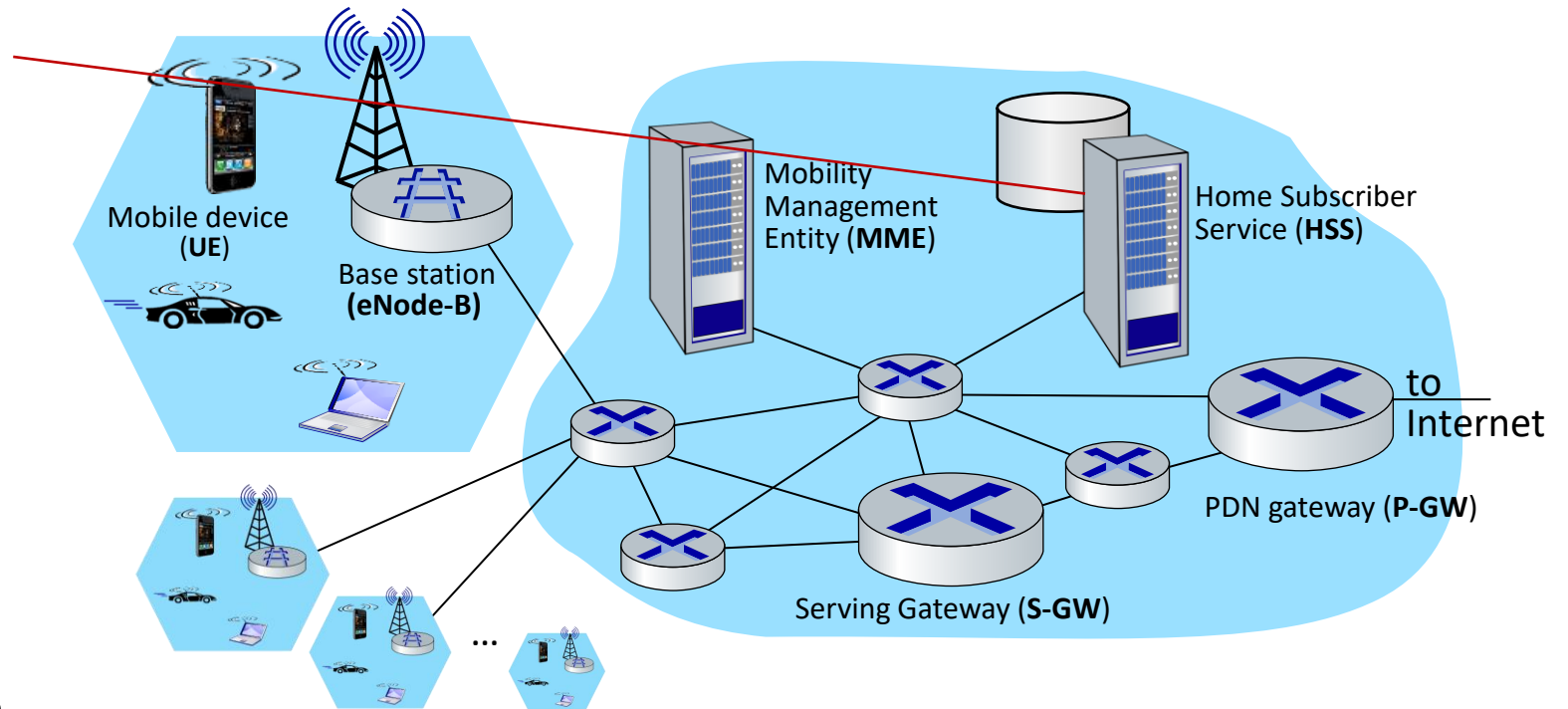
- at “edge” of carrier’s network
- manages wireless radio resources, mobile devices in its coverage area (“cell”)
- coordinates device authentication with other elements
- similar to WiFi AP but:
 - active role in user mobility
 - coordinates with nearby base stations to optimize radio use
- LTE jargon: eNode-B



Elements of 4G LTE architecture

Home Subscriber Service (HSS)

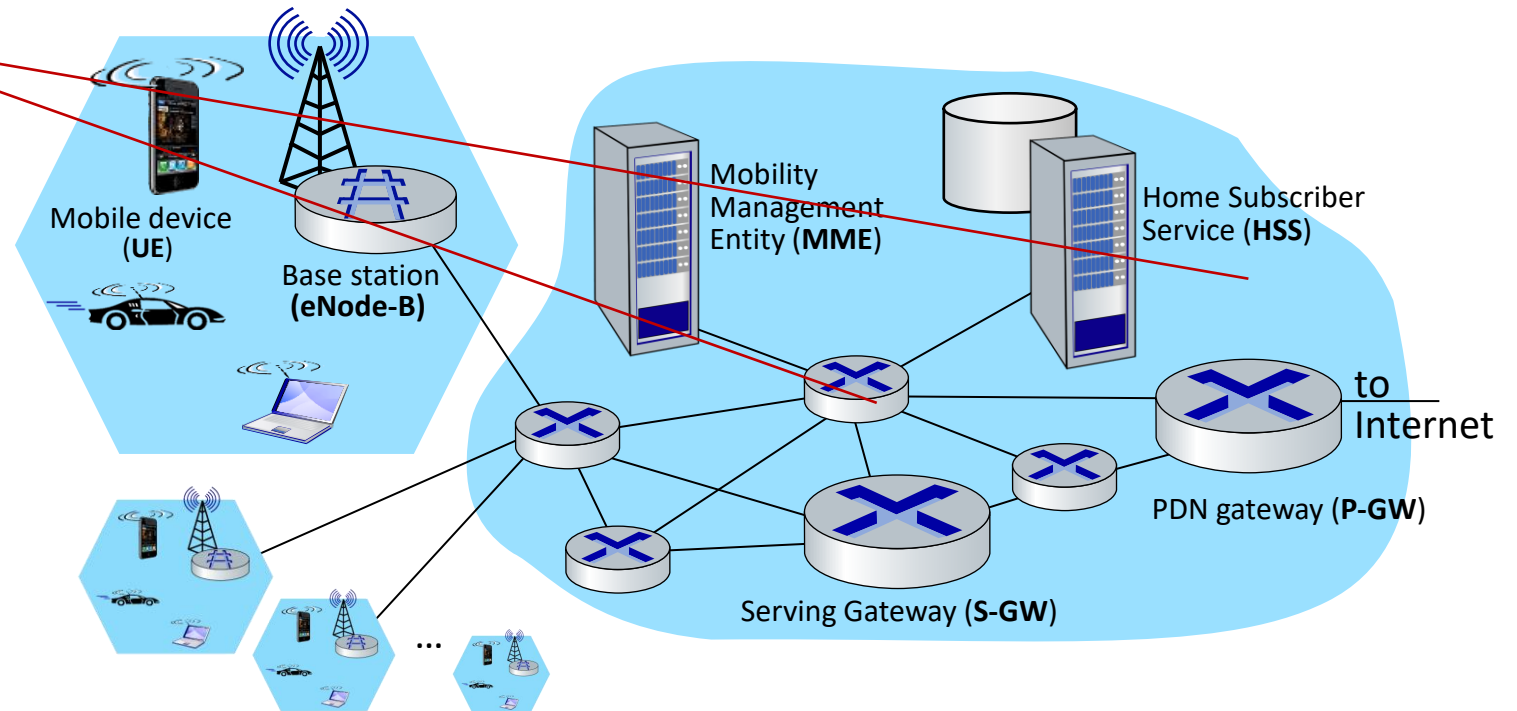
- stores info about mobile devices for which the network of Home Subscriber Service (HSS) is the “home network”
- works with Mobility Management Entity (MME) in device authentication



Elements of 4G LTE architecture

Serving Gateway (S-GW), PDN Gateway (P-GW)

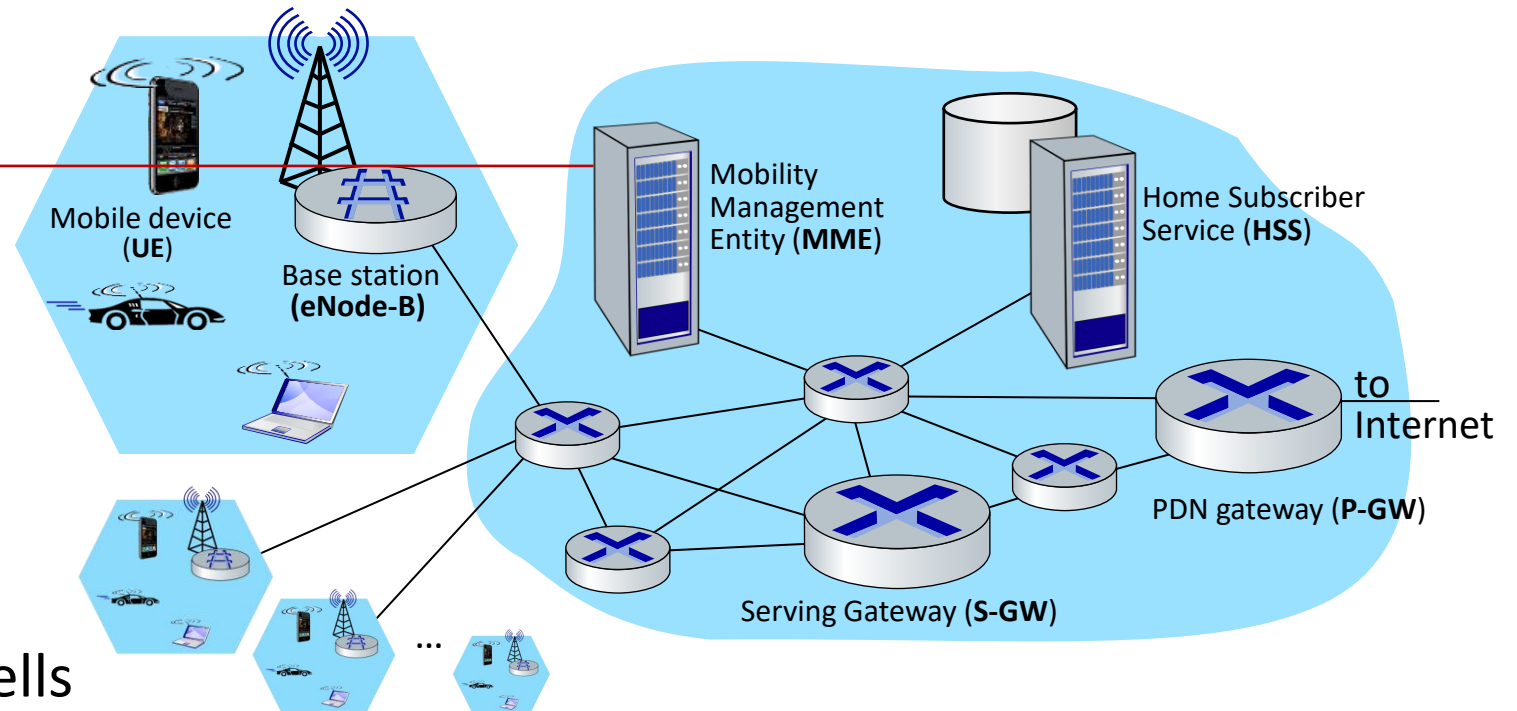
- S-GW
 - gateway to connect eNodeBs and forward a UE's packets as a router
- P-GW
 - gateway to mobile cellular network
 - Looks like other Internet gateway router
 - provides NAT services
- other routers:
 - extensive use of tunneling



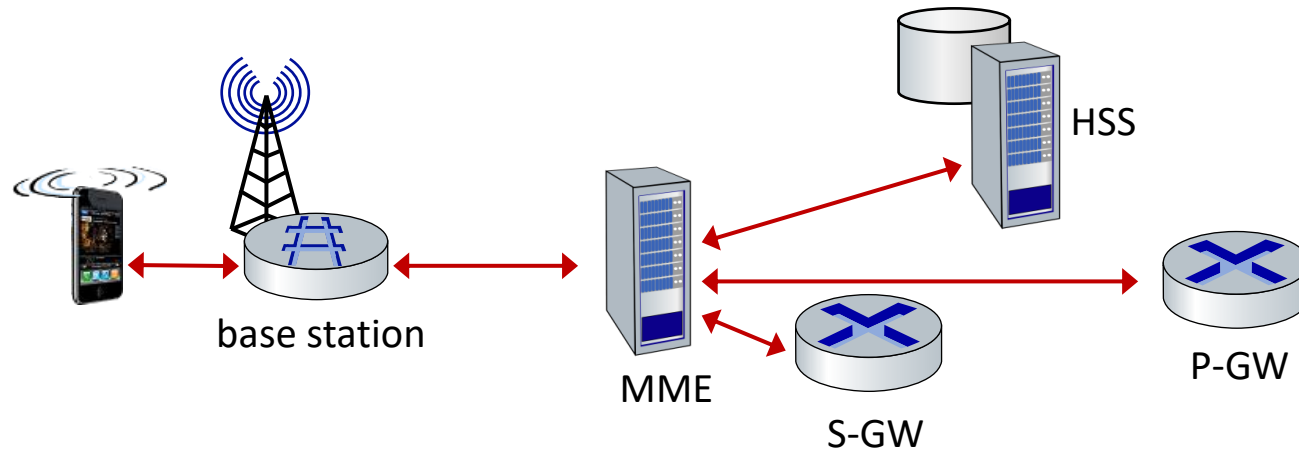
Elements of 4G LTE architecture

Mobility Management Entity

- device authentication (device-to-network, network-to-device) coordinated with mobile home network HSS
- mobile device management:
 - device **handover** between cells
 - tracking/paging device location
- path (tunneling) setup from mobile device to P-GW

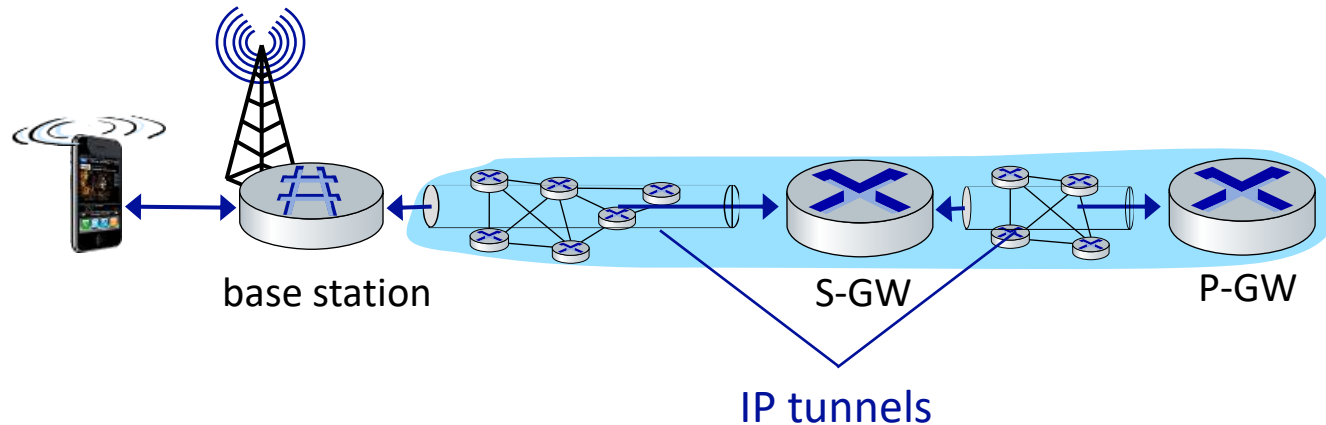


LTE: data plane control plane separation



control plane

- new protocols for mobility management, security, authentication (later)

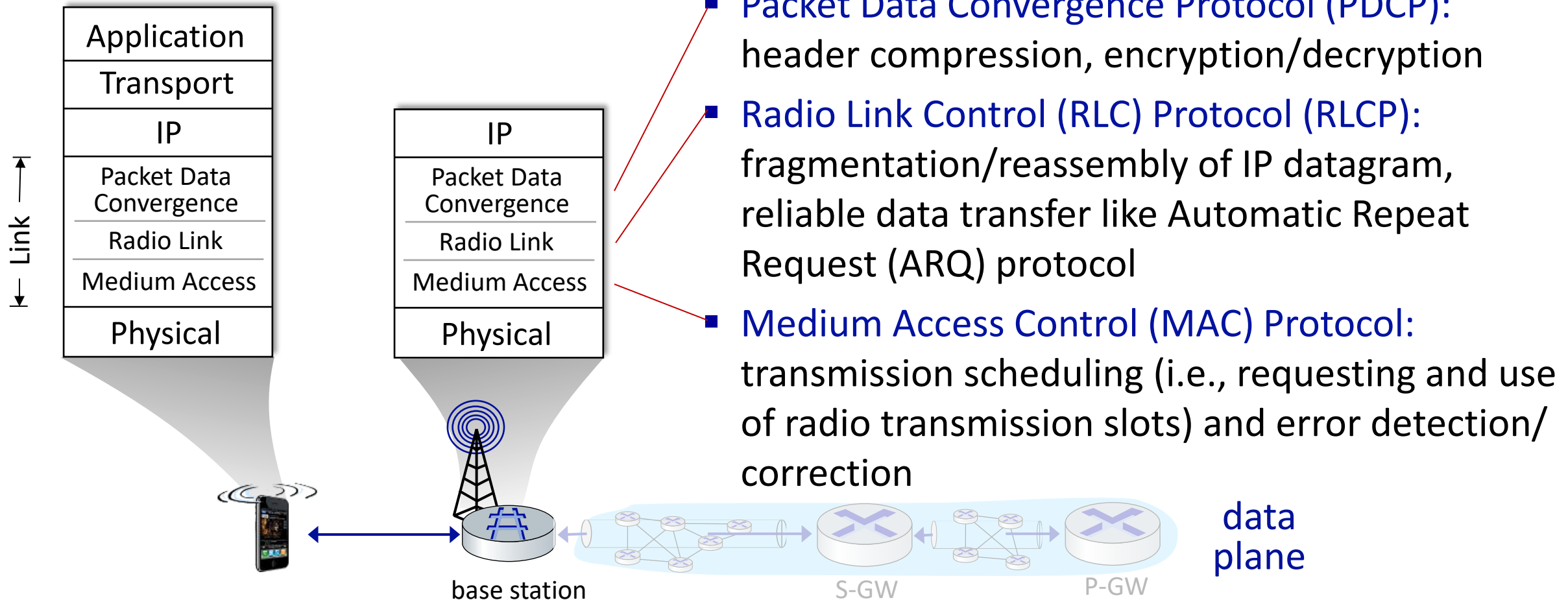


data plane

- new protocols at link, physical layers
- extensive use of tunneling to facilitate mobility

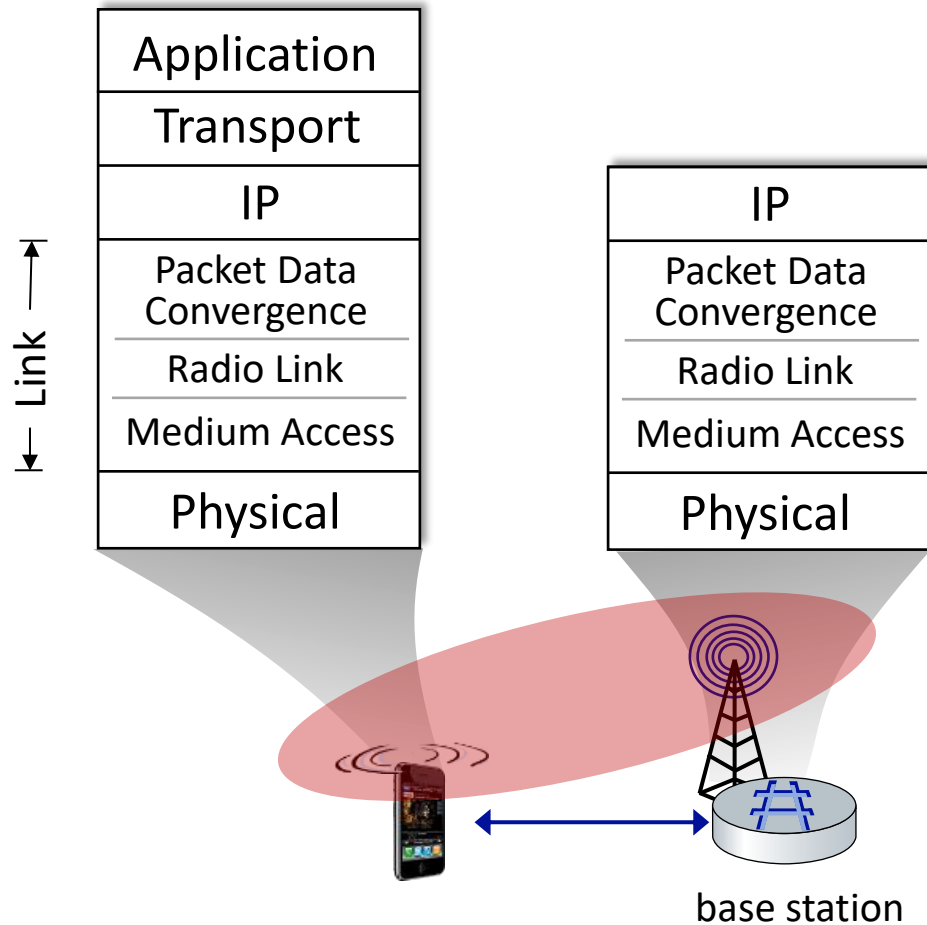
LTE data plane protocol stack: first hop

LTE link layer protocol
(three sublayer protocols):



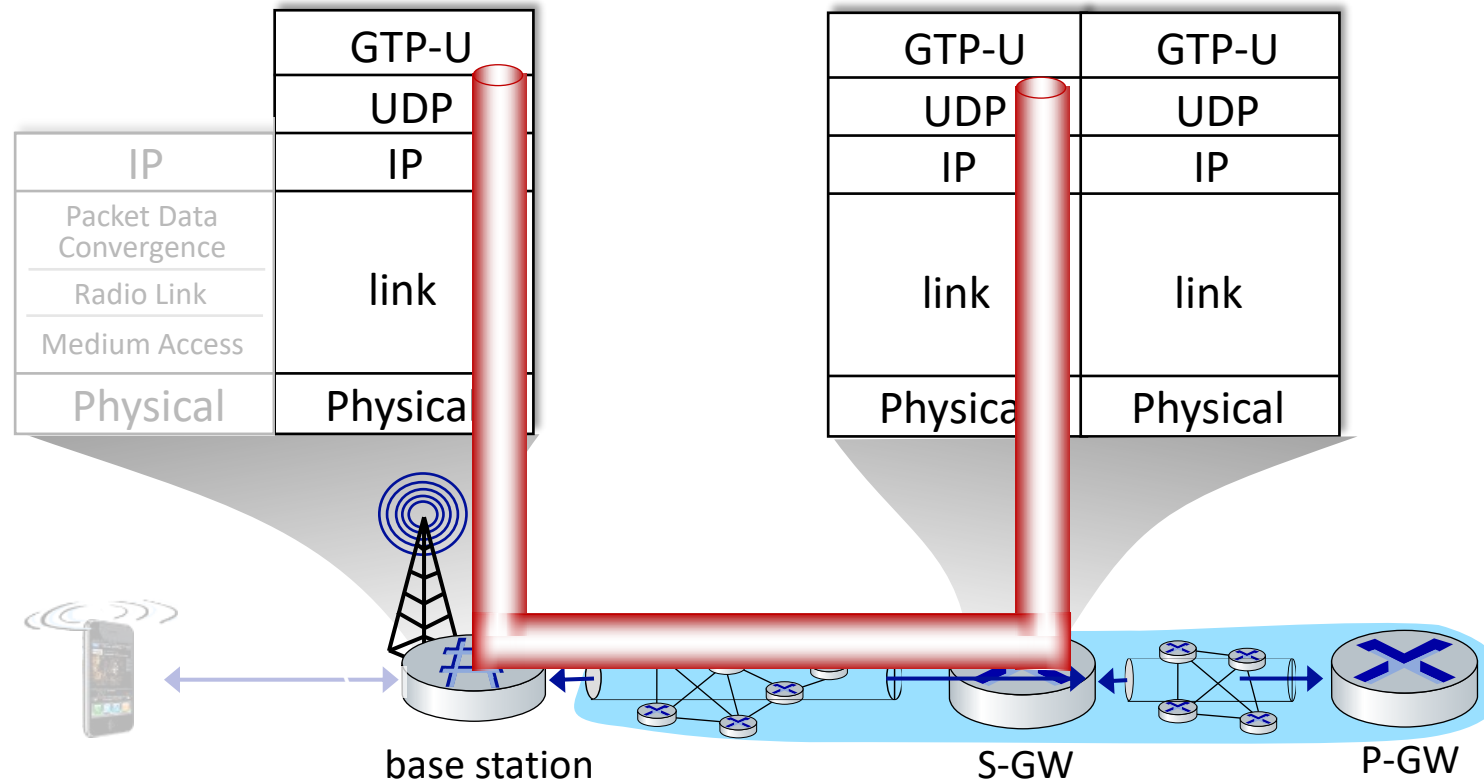
LTE data plane protocol stack: first hop

LTE Radio Access Network (RAN):



- **Downstream channel:** A combination of FDM and TDM within frequency channel (a.k.a. OFDM - Orthogonal Frequency Division Multiplexing)
 - “orthogonal”: minimal interference between channels
- **Upstream channel:** A combination of FDM and TDM similar to OFDM
- each active mobile device allocated two or more 0.5 ms time slots over 12 frequencies
 - scheduling algorithm not standardized – up to operator
 - 100's Mbps per device possible

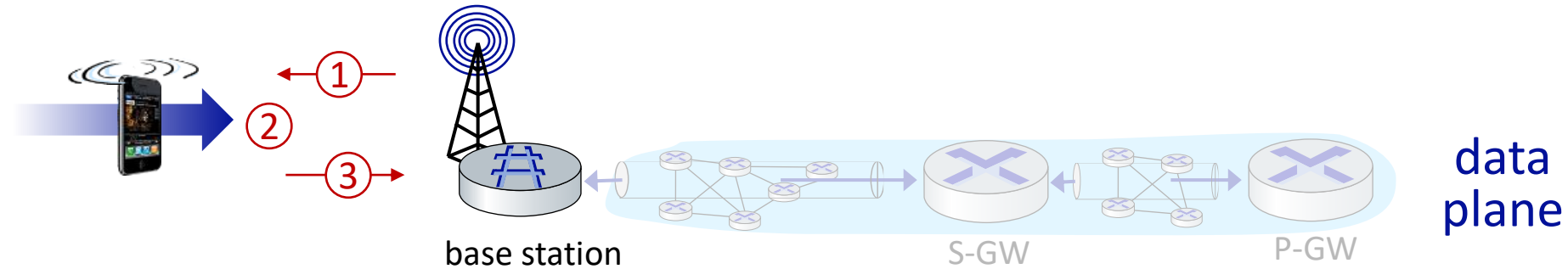
LTE data plane protocol stack: packet core



tunneling:

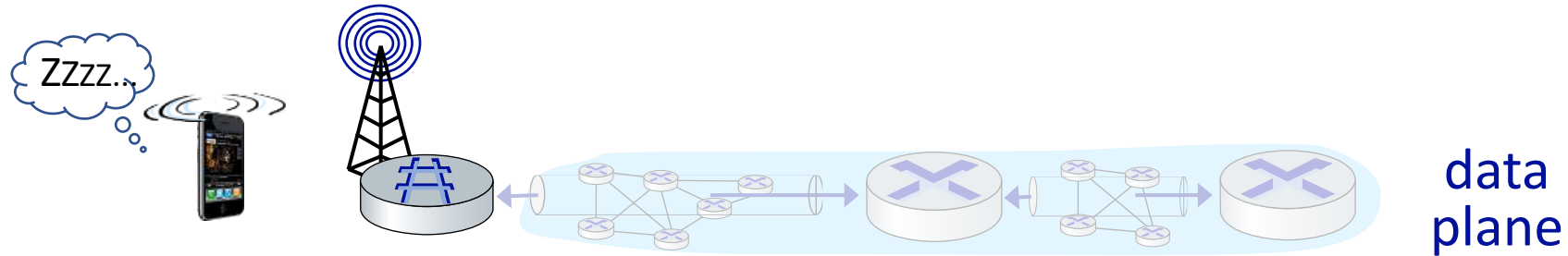
- mobile datagram encapsulated using GPRS Tunneling Protocol (GTP), sent inside UDP datagram to S-GW
 - GPRS: General Packet Radio Service
- S-GW re-tunnels datagrams to P-GW
- supporting mobility: only tunneling endpoints change when mobile user moves

LTE data plane: associating with a BS



- ① BS broadcasts primary synch signal every 5 ms on all frequencies
 - BSs from multiple carriers may be broadcasting synch signals
- ② mobile finds a primary synch signal, then locates 2nd synch signal on this freq.
 - mobile then finds info broadcast by BS: channel bandwidth, configurations; BS's cellular carrier info
 - mobile may get info from multiple base stations, multiple cellular networks
- ③ mobile selects which BS to associate with (*e.g.*, preference for home carrier)
- ④ more steps still needed to authenticate, establish state, set up data plane

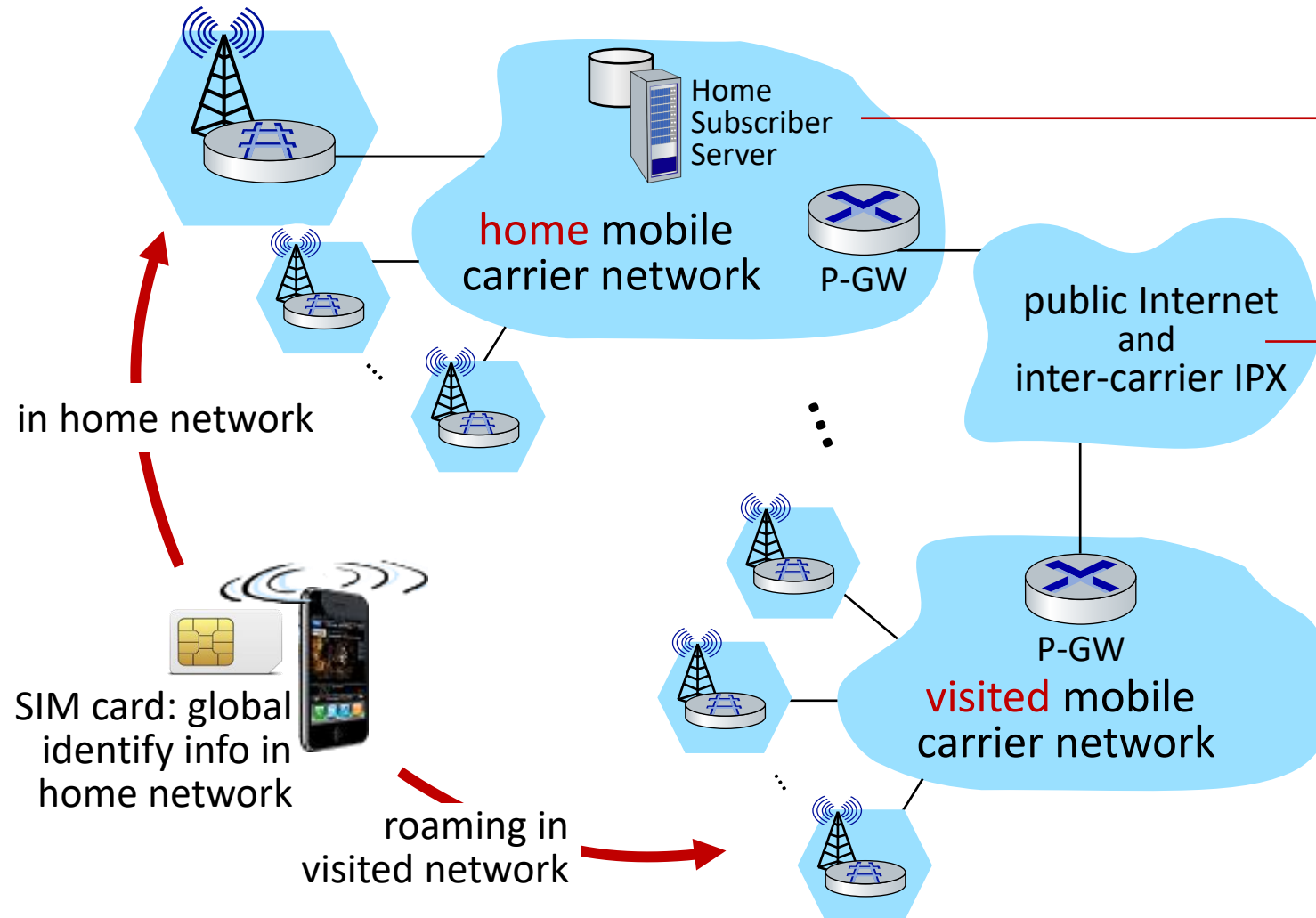
LTE mobiles: sleep modes



as in WiFi, Bluetooth: LTE mobile may put radio to “sleep” to conserve battery:

- **light sleep:** after 100's msec of inactivity
 - wake up periodically (100's msec) to check for downstream transmissions
- **deep sleep:** after 5-10 secs of inactivity
 - mobile may change cells while deep sleeping – need to re-establish association

Global cellular network: a network of IP networks



home network HSS:

- identify & services info, while in home network and roaming

all IP:

- carriers interconnect with each other, and public internet at exchange points
- legacy 2G, 3G: not all IP, handled otherwise

On to 5G!

- **goal:** 10x increase in peak bitrate, 10x decrease in latency, 100x increase in traffic capacity over 4G
- **5G NR (new radio):**
 - two frequency bands: FR1 (450 MHz–6 GHz) and FR2 (24 GHz–52 GHz): millimeter wave frequencies
 - not backwards-compatible with 4G
 - MIMO: multiple directional antennae
- **millimeter wave frequencies:** much higher data rates, but over shorter distances
 - pico-cells: cells diameters: 10-100 m
 - massive, dense deployment of new base stations required

5G

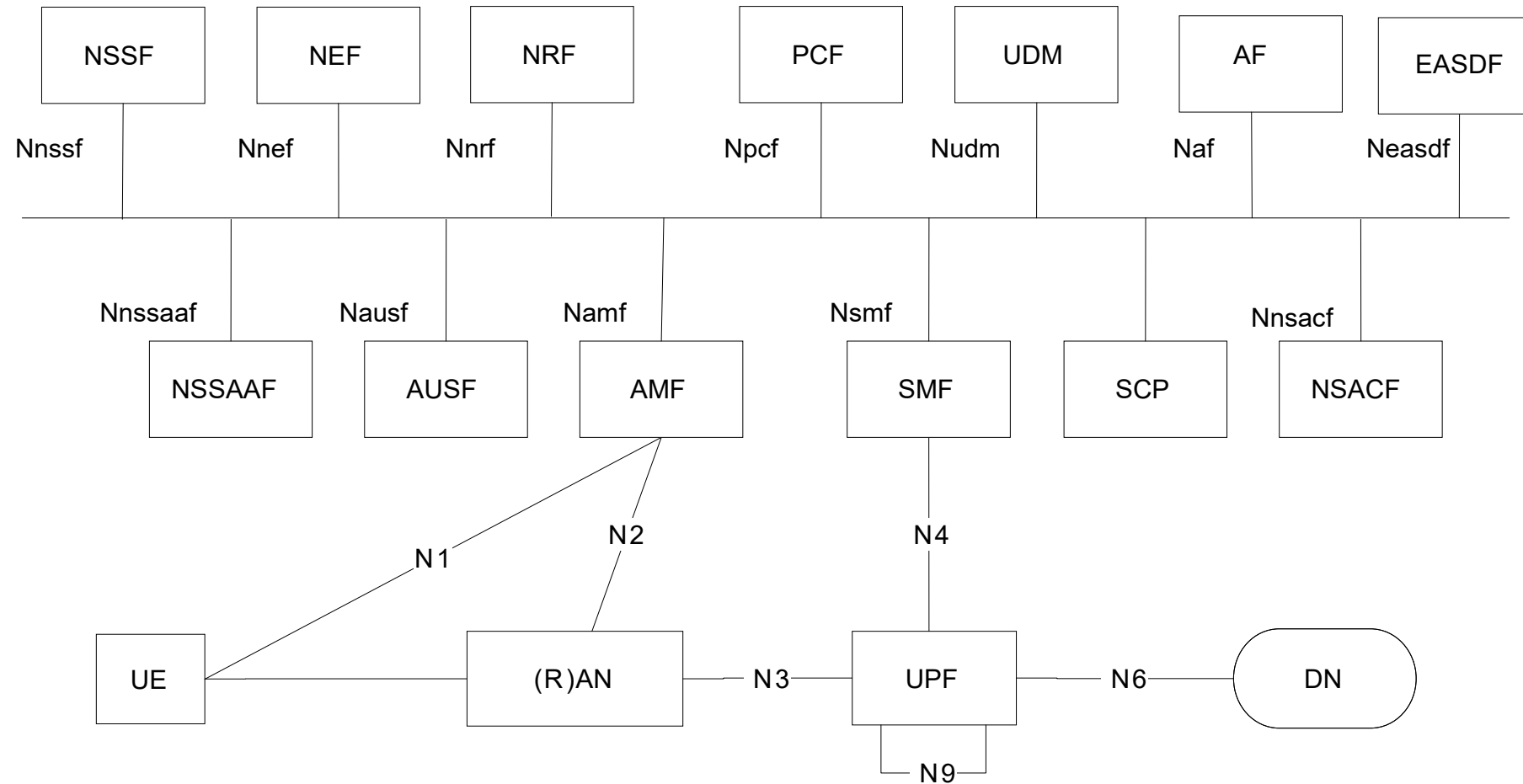
[Source] <https://www.3gpp.org/technologies/5g-system-overview>

- 5G on top of Softwarization and Virtualization
 - 5G networks are constructed by Network Functions Virtualization (NFV).
- Release 15: The Specification of 5G Phase 1
 - This introduces a New Radio (NR) transmission technique.
 - Other key concepts are defined such as an industry-grade reliability, an extended modularity, or a faster response time.
- 5G Services
 - Enhanced Mobile Broadband (eMBB)
 - Critical Communications (CC) and Ultra Reliable and Low Latency Communications (URLLC)
 - Massive Machine Type Communications (mMTC) or Massive Internet of Things (mIoT)
 - Flexible network operations.

5G Architecture

5G System Overview

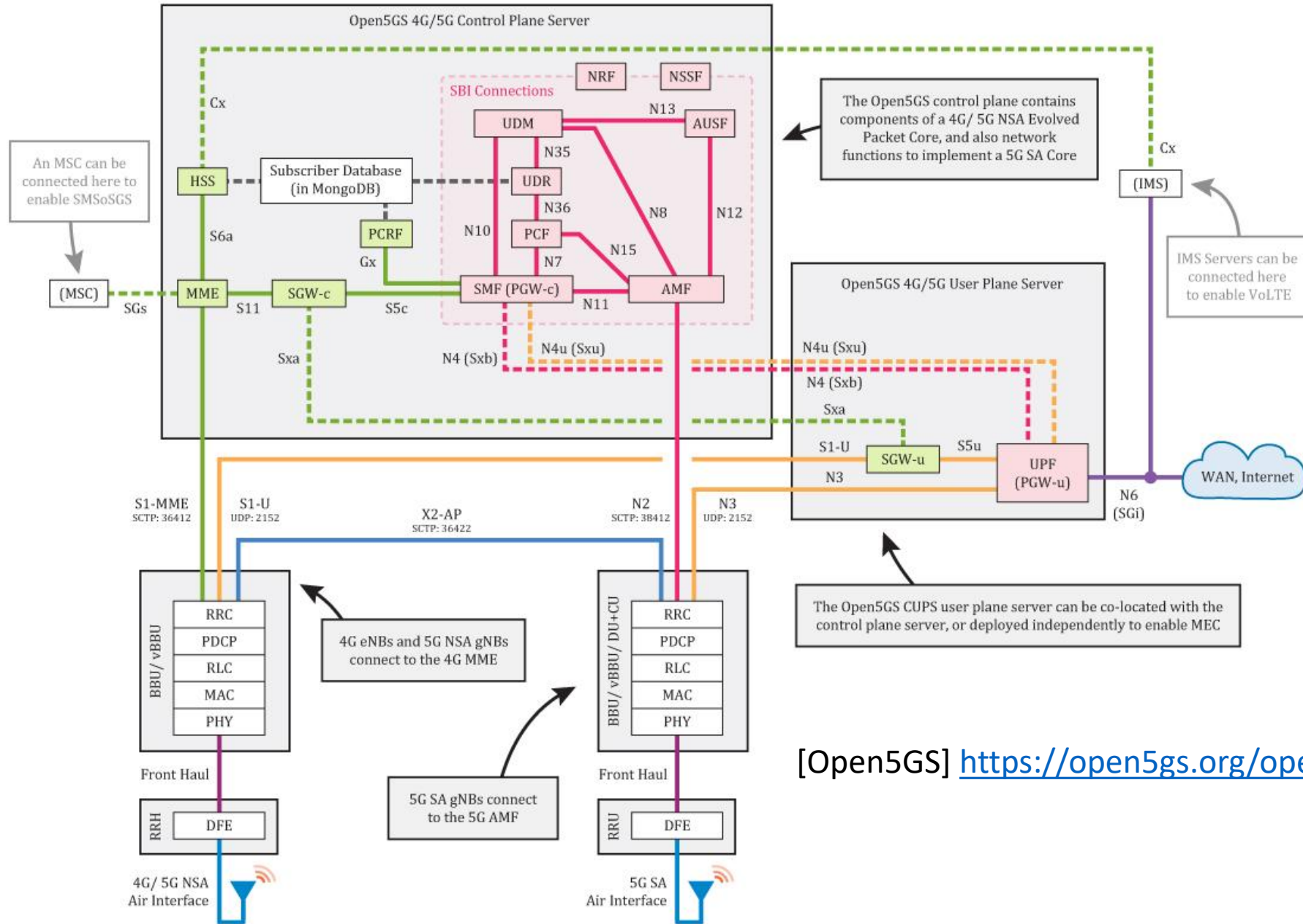
- <https://www.3gpp.org/technologies/5g-system-overview>



[3GPP TS 23.501] System architecture for the 5G System (5GS)

<https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3144>

Open5GS



[Open5GS] <https://open5gs.org/open5gs/docs/guide/01-quickstart/>

Chapter 7 outline

- Introduction

Wireless

- Wireless links and network characteristics
- WiFi: 802.11 wireless LANs
- Cellular networks: 4G and 5G



Mobility

- Mobility management: principles
- Mobility management: practice
 - 4G/5G networks
 - Mobile IP
- Mobility: impact on higher-layer protocols

What is mobility?

- spectrum of mobility, from the **network** perspective:

no mobility

high mobility



device moves
between
networks, but
powers down
while moving

device moves
within same AP in
one provider
network

device moves
among APs in
one provider
network

device moves
among multiple
provider networks,
while maintaining
ongoing
connections

We're interested in these!

Mobility approaches

- **let network (routers) handle it:**
 - routers advertise well-known name, address (e.g., permanent 32-bit IP address), or number (e.g., cell #) of visiting mobile node via usual routing table exchange
 - Internet routing could do this already *with no* changes! Routing tables indicate where each mobile located via longest prefix match!

Mobility approaches

- let network (routers) handle it:
 - routers advertise well-known address (e.g., permanent 32-bit IP address), or number of visiting mobile node via usual routing table exchange
 - Internet routing could do it *very easily with no changes!* Routing tables indicate where each mobile located via longest prefix match!
- **let end-systems handle it: functionality at the “edge”**
 - *indirect routing*: communication from correspondent to mobile goes through home network, then forwarded to remote mobile
 - *direct routing*: correspondent gets foreign address of mobile, send directly to mobile

not
scalable
to billions of
mobiles

Contacting a mobile friend:

Consider friend frequently changing locations, how do you find him/her?

- search all phone books?
- expect her to let you know where he/she is?

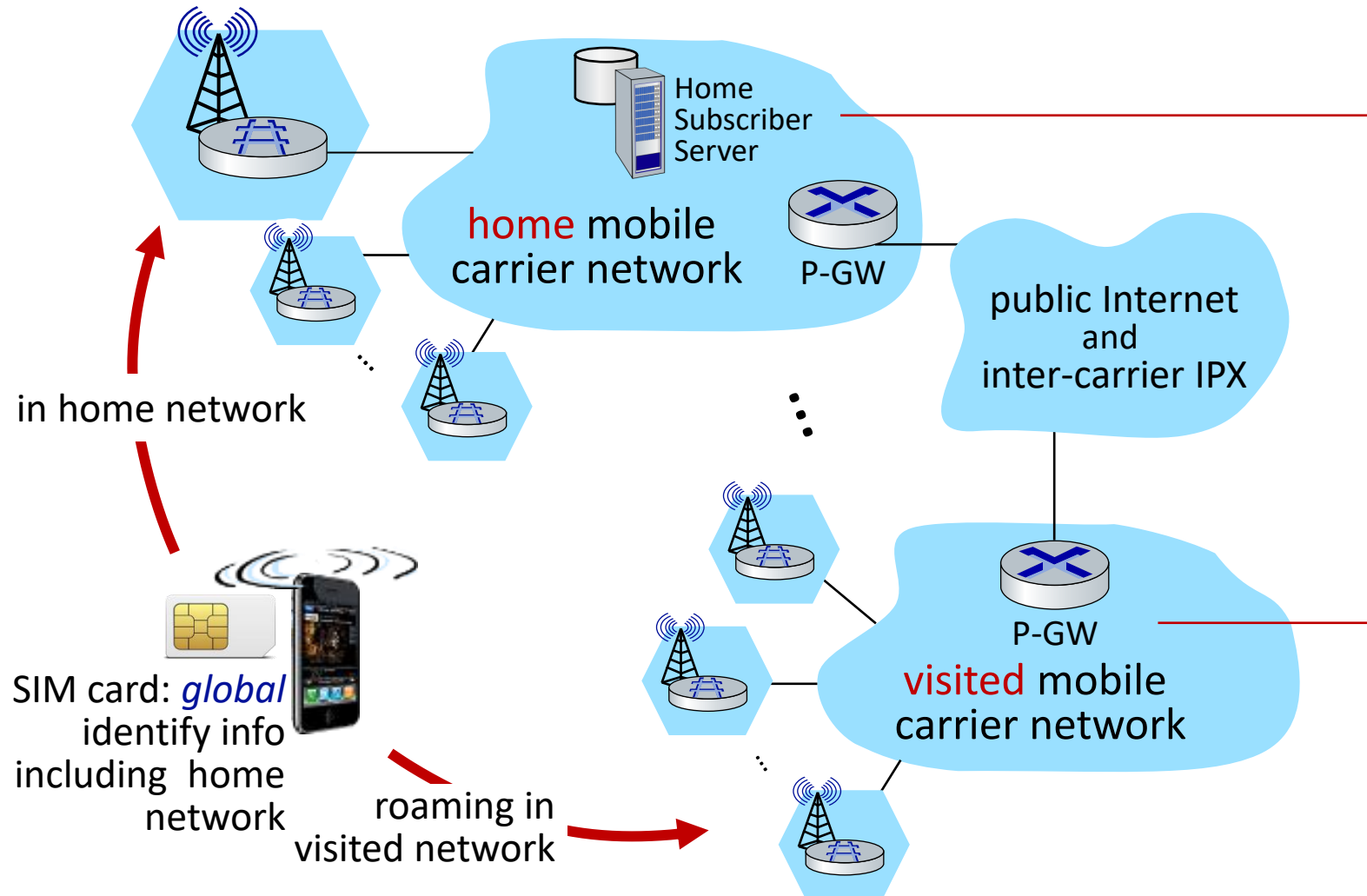
- call his/her parents?
- Facebook!

The importance of having a “home”:

- a definitive source of information about you
- a place where people can find out where you are



Home network, visited network: 4G/5G



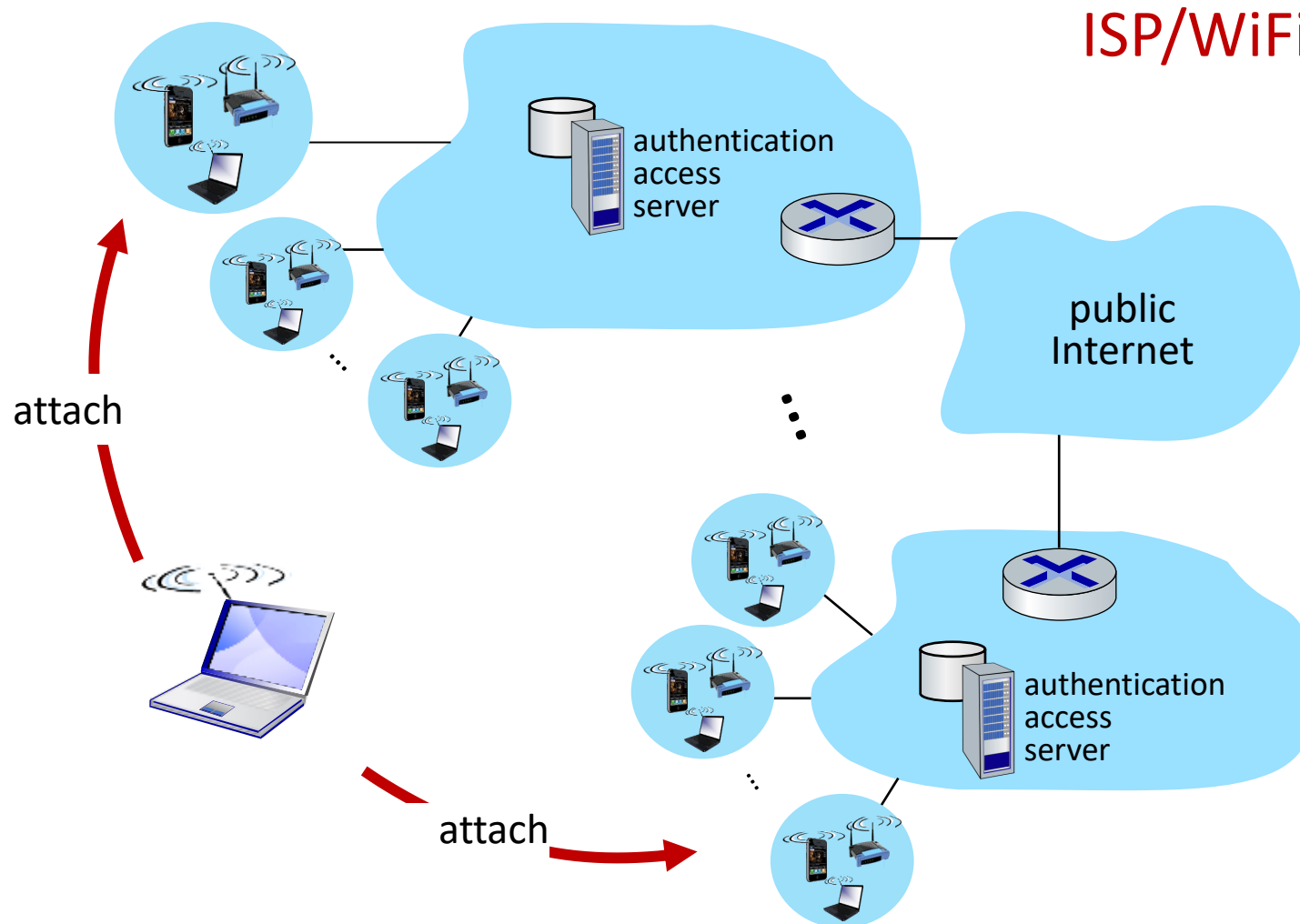
home network:

- (paid) service plan with cellular provider, e.g., Verizon, Orange
- home network HSS stores identify & services info

visited network:

- any network other than your home network
- service agreement with other networks: to provide access to visiting mobile

Home network, visited network: ISP/WiFi



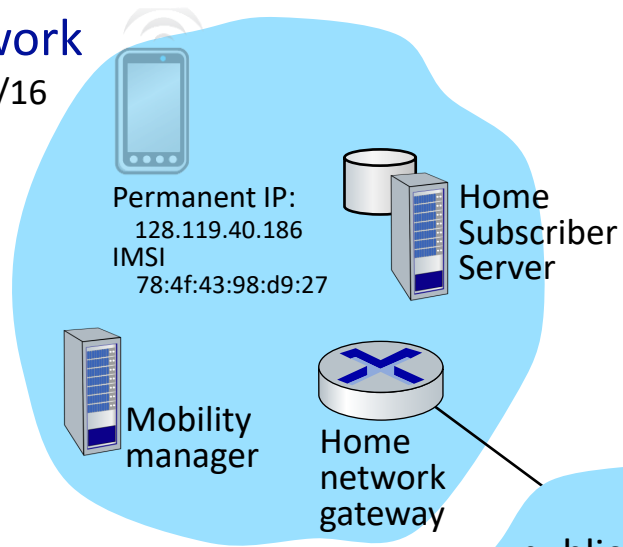
ISP/WiFi: no notion of global “home”

- credentials from ISP (e.g., username, password) stored on device or with user
- ISPs may have national, international presence
- different networks: different credentials
 - some exceptions (e.g., eduroam)
 - architectures exist (mobile IP) for 4G-like mobility, but not used

Home network, visited network: generic

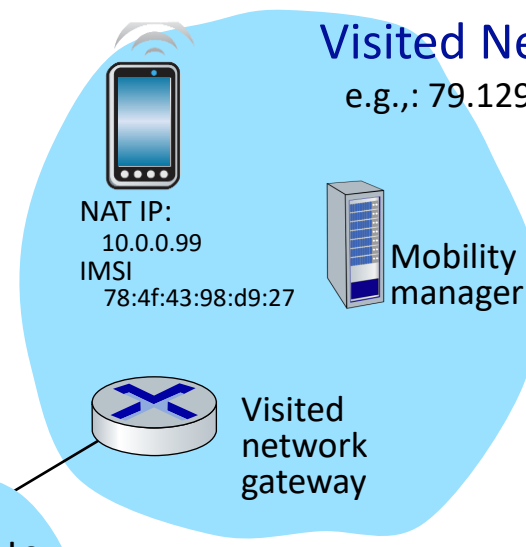
Home Network

e.g.,: 128.119/16



Visited Network

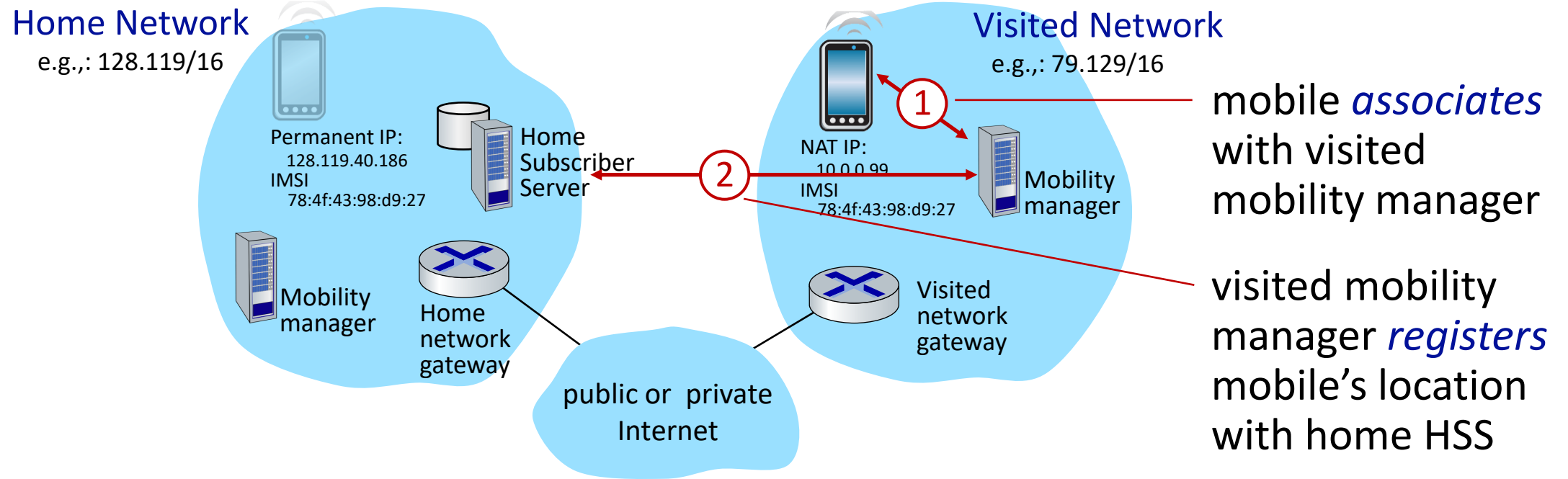
e.g.,: 79.129/16



public or private
Internet

Correspondent

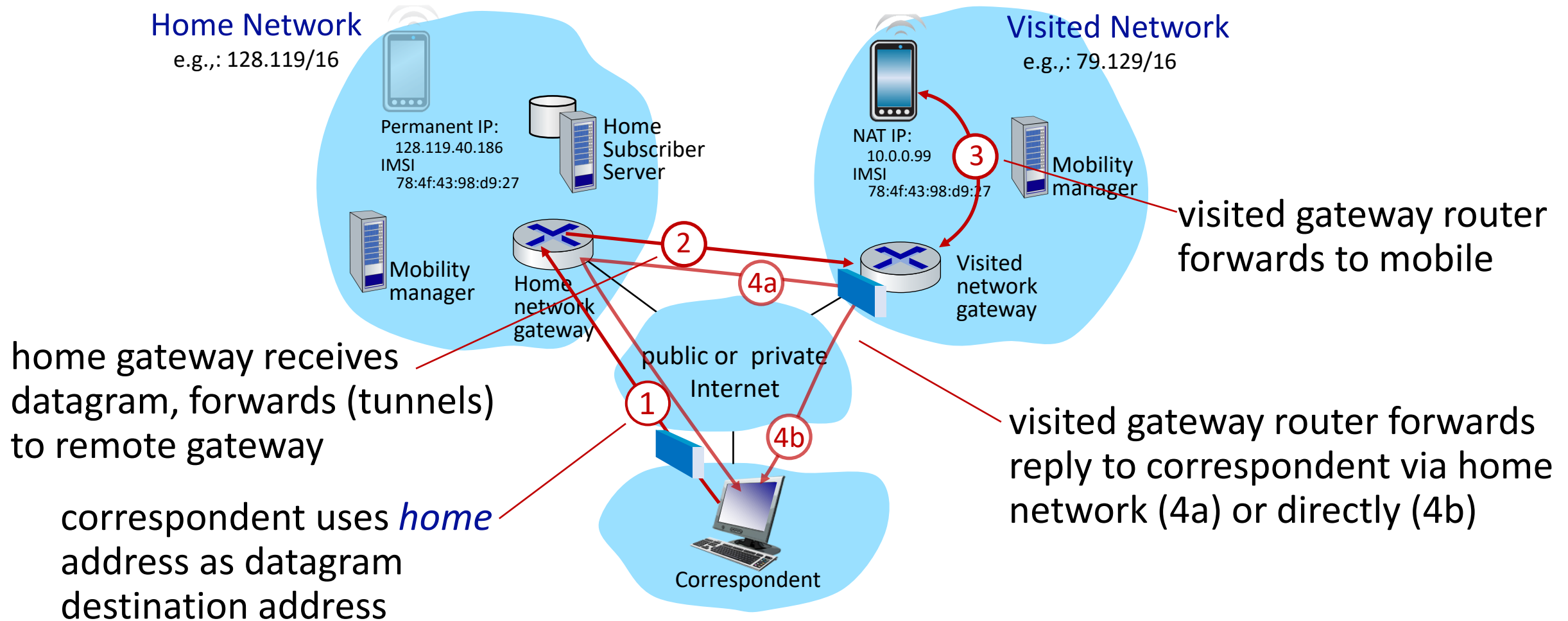
Registration: home needs to know where you are!



end result:

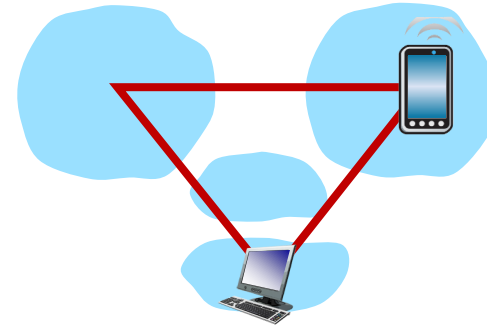
- visited mobility manager knows about mobile
- home HSS knows location of mobile

Mobility with indirect routing

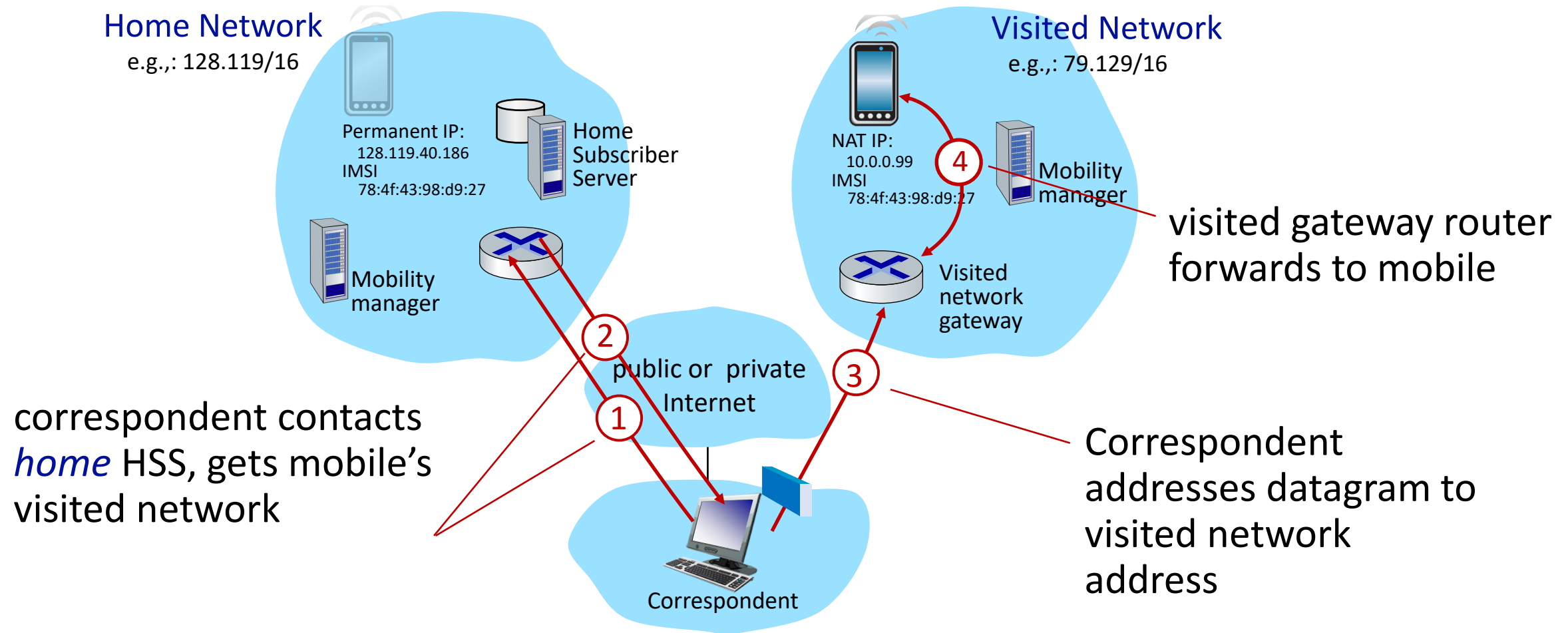


Mobility with indirect routing: comments

- triangle routing:
 - inefficient when correspondent and mobile are in same network
- mobile moves among visited networks: transparent to correspondent!
 - registers in new visited network
 - new visited network registers with home HSS
 - datagrams continue to be forwarded from home network to mobile in new network
 - *on-going (e.g., TCP) connections between correspondent and mobile can be maintained!*



Mobility with direct routing



Mobility with direct routing: comments

- overcomes triangle routing inefficiencies
- *non-transparent to correspondent*: correspondent must get care-of-address from home agent
- what if mobile changes visited network?
 - can be handled, but with additional complexity

Chapter 7 outline

- Introduction

Wireless

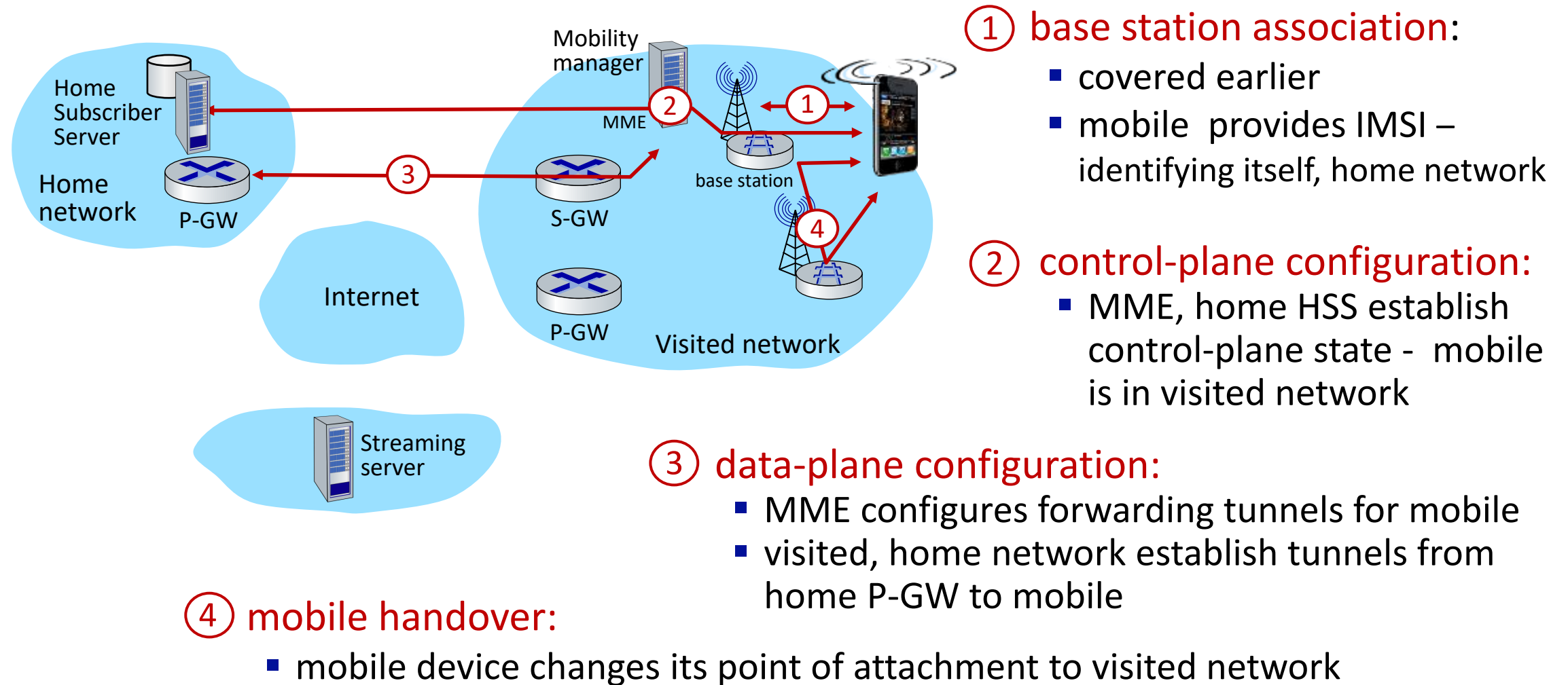
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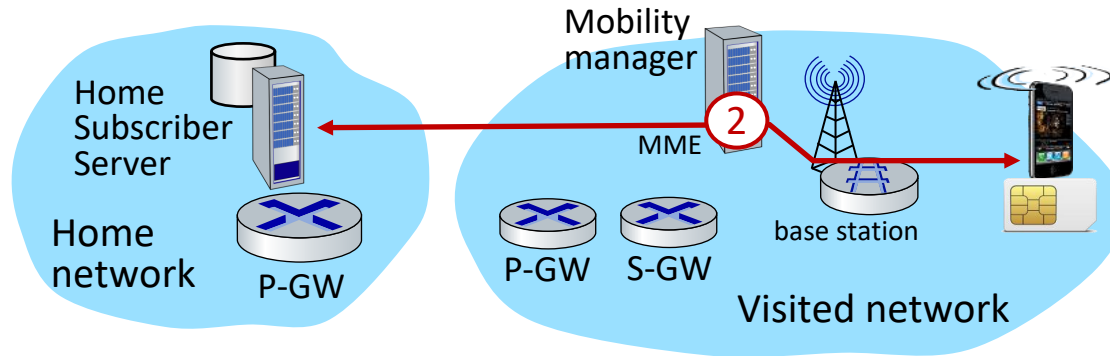
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Mobility in 4G networks: major mobility tasks



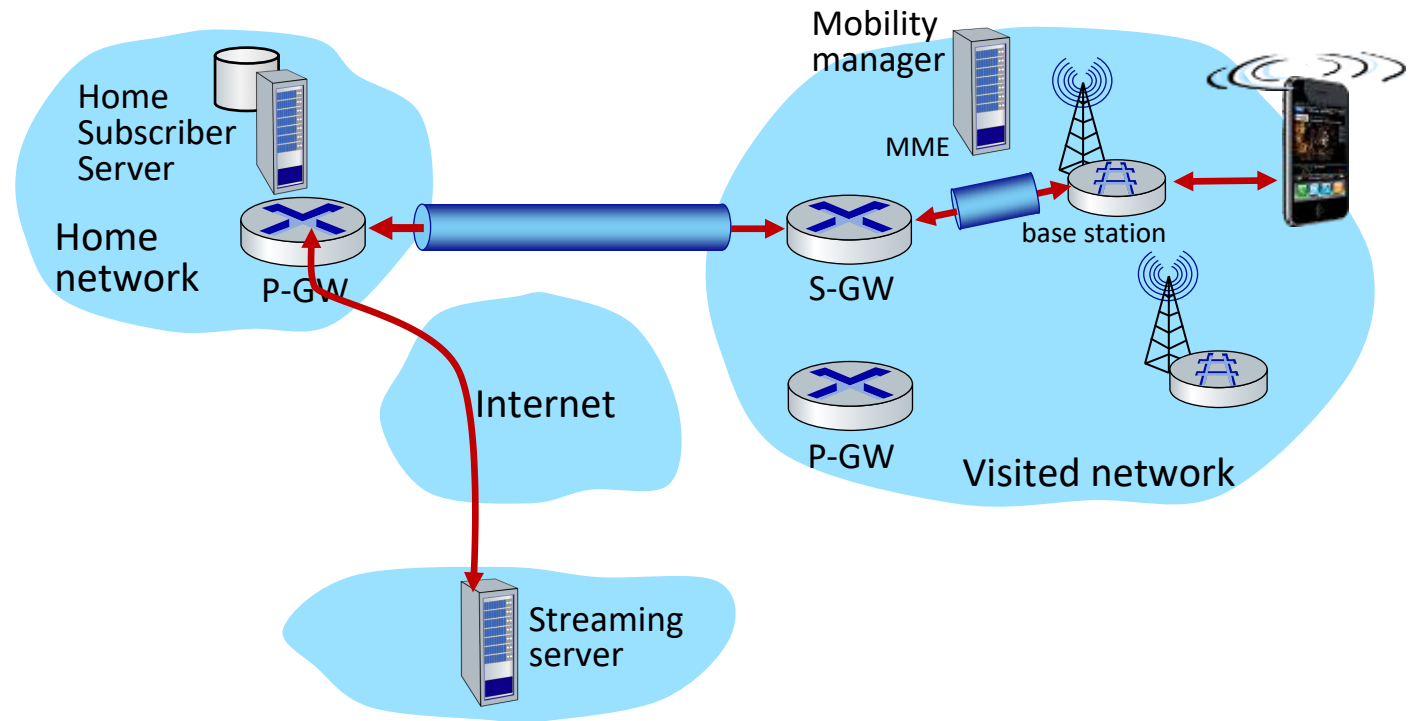
Configuring LTE control-plane elements



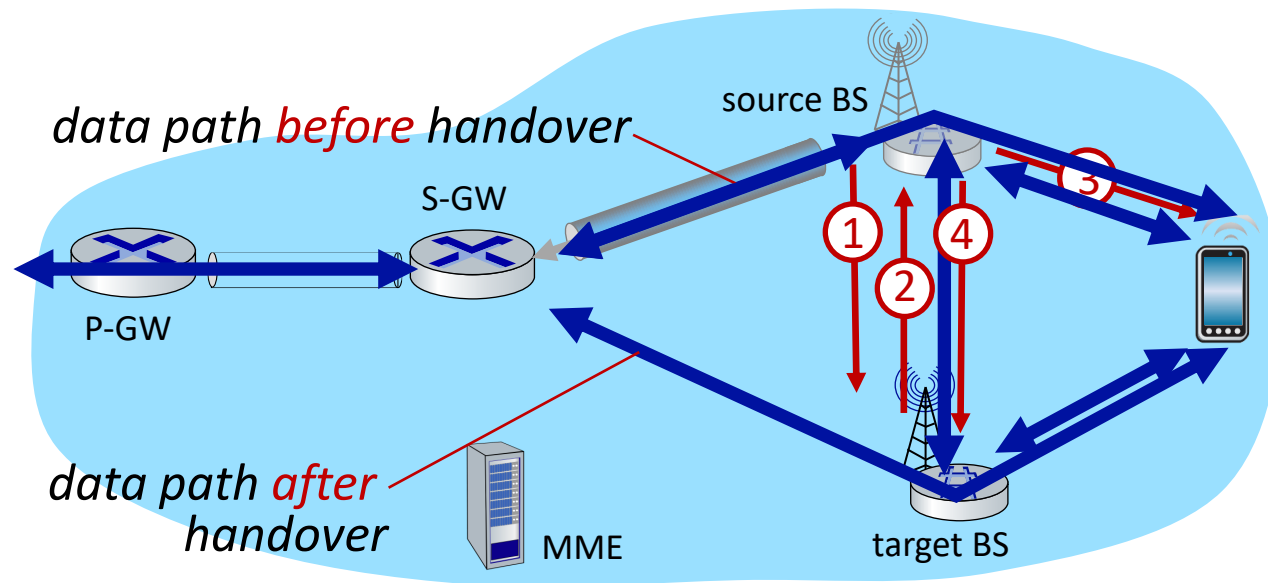
- Mobile communicates with local MME via BS control-plane channel
- MME uses mobile's IMSI info to contact mobile's home HSS
 - retrieve authentication, encryption, network service information
 - home HSS knows mobile now resident in visited network
- BS, mobile select parameters for BS-mobile data-plane radio channel

Configuring data-plane tunnels for mobile

- **S-GW to BS tunnel:** when mobile changes base stations, simply change endpoint IP address of tunnel
- **S-GW to home P-GW tunnel:** implementation of indirect routing
- **tunneling via GTP** (GPRS tunneling protocol): mobile's datagram to streaming server encapsulated using GTP inside UDP, inside datagram



Handover between BSs in same cellular network



① current (source) BS selects target BS, sends *Handover Request* message to target BS

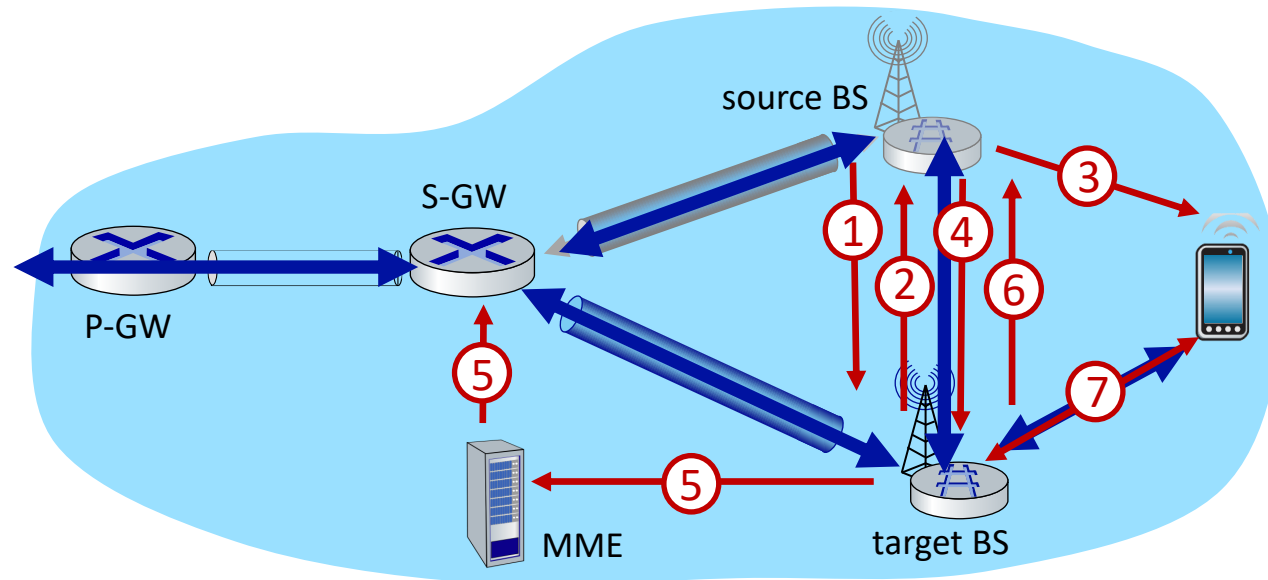
② target BS pre-allocates radio time slots, responds with HR ACK with info for mobile

③ source BS informs mobile of new BS

- mobile can now send via new BS - handover *looks* complete to mobile

④ source BS stops sending datagrams to mobile, instead forwards to new BS (who forwards to mobile over radio channel)

Handover between BSs in same cellular network



- ⑤ target BS informs MME that it is new BS for mobile
- MME instructs S-GW to change tunnel endpoint to be (new) target BS

- ⑥ target BS ACKs back to source BS: handover complete, source BS can release resources

- ⑦ mobile's datagrams now flow through new tunnel from target BS to S-GW

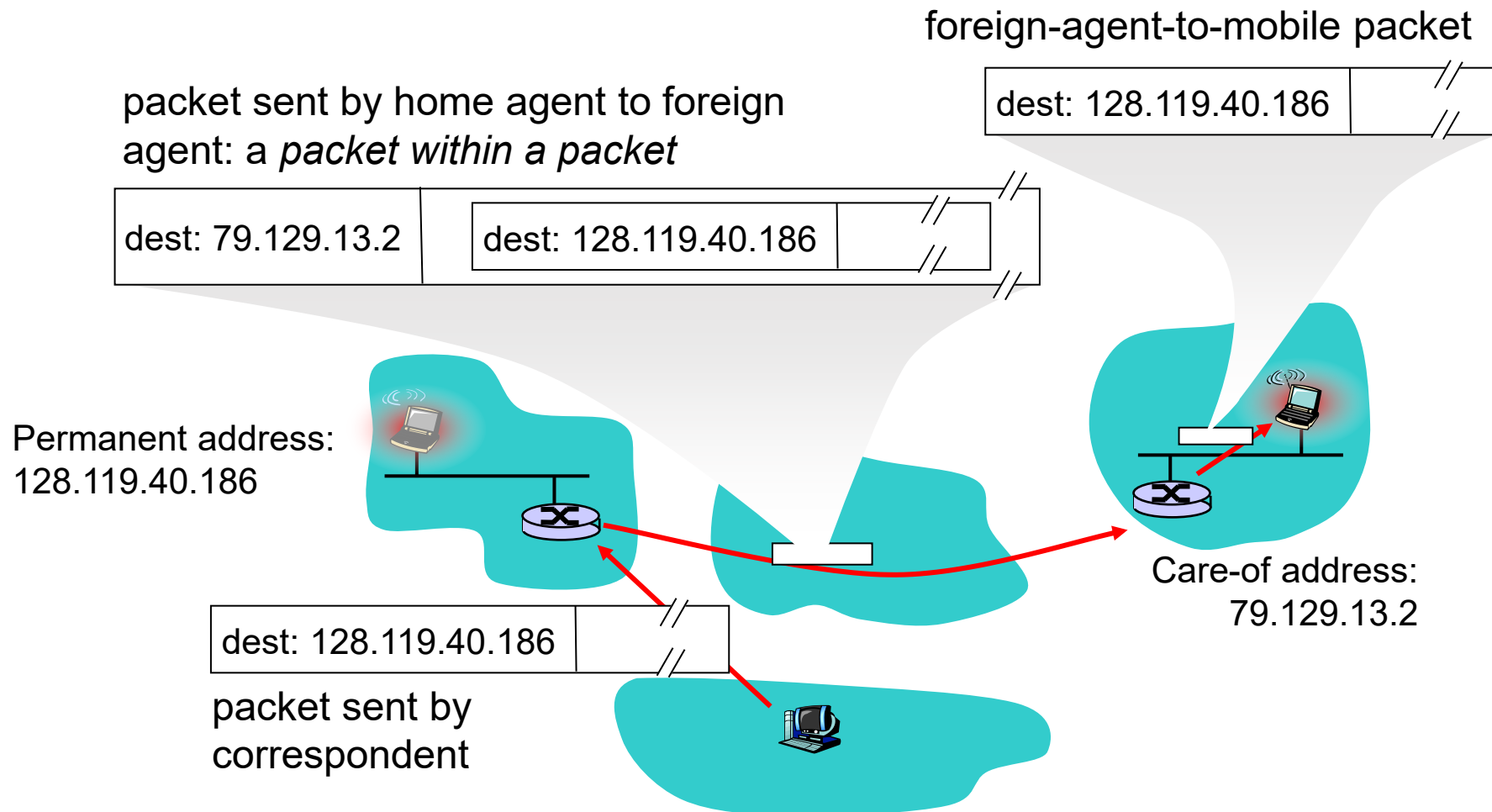
Mobile IP (1): Introduction

- mobile IP architecture standardized ~20 years ago [RFC 5944]
 - long before ubiquitous smartphones, 4G support for Internet protocols
 - did not see wide deployment/use
 - perhaps WiFi for Internet, and 2G/3G phones for voice were “good enough” at the time
- mobile IP architecture:
 - indirect routing to node (via home network) using tunnels
 - mobile IP home agent: combined roles of 4G HSS and home P-GW
 - mobile IP foreign agent: combined roles of 4G MME and S-GW
 - protocols for agent discovery in visited network, registration of visited location in home network via ICMP extensions

Mobile IP (2): Original Mobile IP

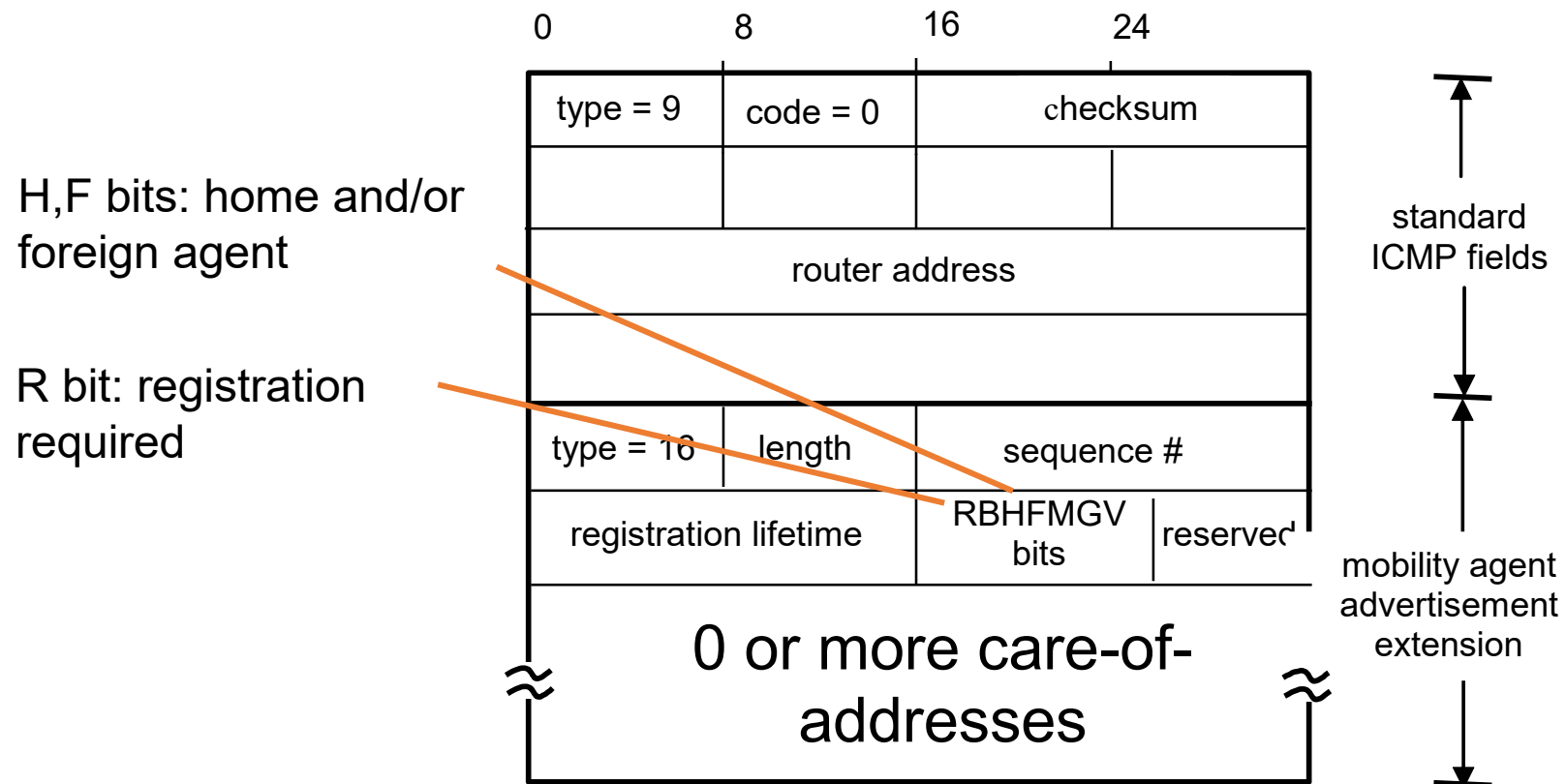
- RFC 3344: IP Mobility Support for IPv4
 - URL: <https://www.rfc-editor.org/rfc/rfc3344>
 - Revision (RFC 5944): <https://www.rfc-editor.org/rfc/rfc5944>
- has many features we've seen:
 - home agents, foreign agents, foreign-agent registration, care-of-addresses, encapsulation (packet-within-a-packet)
- three components to standard:
 - indirect routing of datagrams
 - agent discovery
 - registration with home agent

Mobile IP (3): Indirect Routing

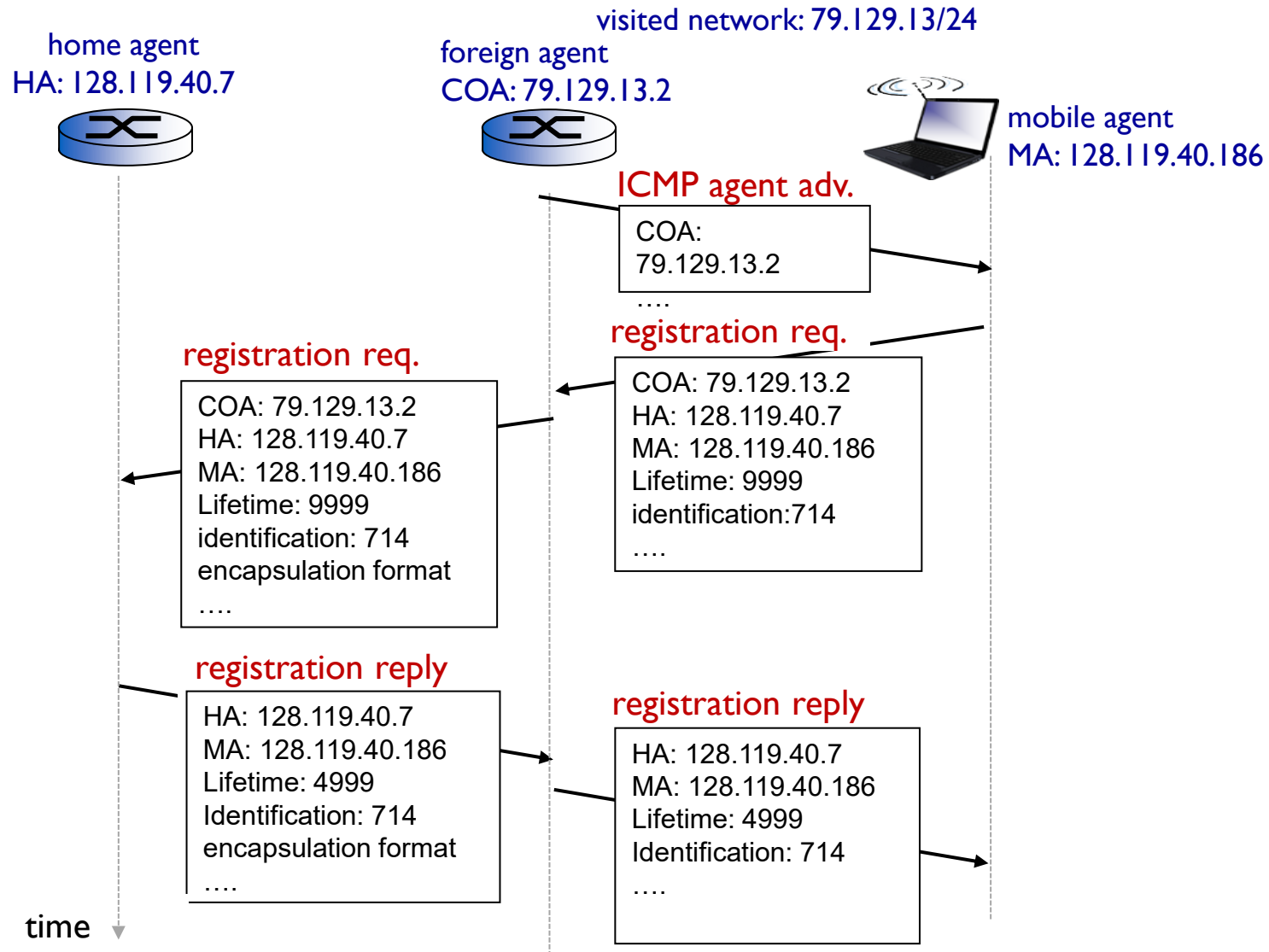


Mobile IP (4): Agent Discovery

- *agent advertisement*: foreign/home agents advertise service by broadcasting ICMP messages (typefield = 9)



Mobile IP (5): Registration Example



Wireless, mobility: impact on higher layer protocols

- logically, impact *should* be minimal ...
 - best effort service model remains unchanged
 - TCP and UDP can (and do) run over wireless, mobile
- ... but performance-wise:
 - packet loss/delay due to bit-errors (discarded packets, delays for link-layer retransmissions), and handover loss
 - TCP interprets loss as congestion, will decrease congestion window unnecessarily
 - delay impairments for real-time traffic
 - bandwidth a scarce resource for wireless links

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